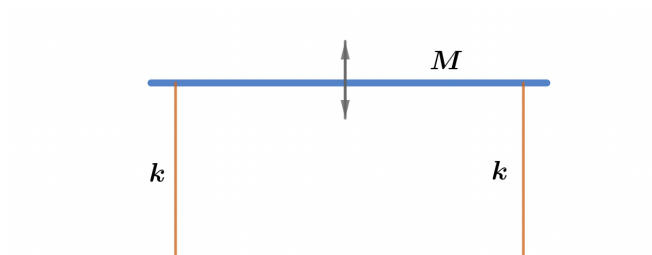


2022B F=ma Exam: Problem 18

Kevin S. Huang



If the rod uniformly displaced downward x below equilibrium, the restoring force is

$$F = -2kx$$

By Newton's 2nd law,

$$F = M\ddot{x} = -2kx$$

$$\ddot{x} = -\frac{2k}{M}x$$

which corresponds to simple harmonic motion with angular frequency

$$\omega = \sqrt{\frac{2k}{M}}$$

The frequency is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{2k}{M}}$$

so the answer is D.